

MODULE 2

1.

List the components of register transfer logic.

The set of registers in the system and their functions: A register also encompasses all type of registers including shift registers, counters and memory units.

The binary-coded information stored in the registers: The binary information stored in registers may be binary numbers, binary coded decimal numbers, alphanumeric characters, control information or any other binary coded information.

The operations performed on the information stored in the registers: The operations performed on data stored in registers are called micro operations. Examples are shift, count, add, clear and load

The control functions that initiate the sequence of operations: The control functions that initiate the sequence of operations consists of timing signals that sequence the operations one at a time.

2.

What is microoperation?

Micro-operations: operations executed on data stored in one or more registers. For any function of the computer, a sequence of micro-operations is used to describe it

Example: Shift, count, clear, add & load

3.

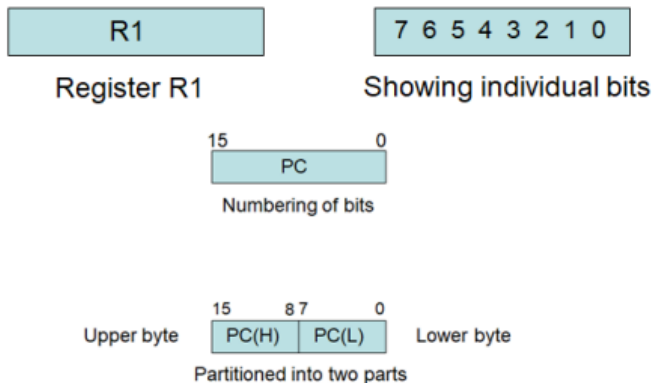
What are the different types of micro operations?

- a. Register transfer microoperations
- b. Arithmetic microoperations (on numeric data stored in the registers)
- c. Logic microoperations (bit manipulations on non-numeric data)
- d. Shift microoperations

4.

How do you represent register in register transfer logic.

- Rectangular box with name of the register inside,
- The individual cells is assigned a letter with a subscript number,
- The numbering of cells from right to left can be marked on top of the box,
- 16 bit register is partitioned into 2 parts , bits 1 to 8 are assigned the letter L(for low) and bits 9 to 16 are assigned the letter H(for high)



5.

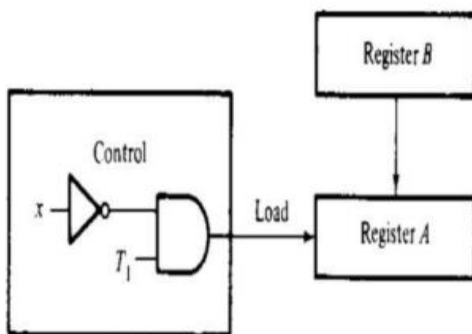
Explain the register transfer micro operation with the help of an example.

- Information transfer from one register to another is described by a *replacement operator*: $R2 \leftarrow R1$
- This statement denotes a transfer of the content of register R1 into register R2
- The transfer happens in one clock cycle
- The content of the R1 (source) does not change
- The content of the R2 (destination) will be lost and replaced by the new data transferred from R1
- We are assuming that the circuits are available from the outputs of the source register to the inputs of the destination register, and that the destination register has a parallel load capability

6.

With a neat diagram explain how the conditional transfer operation is implemented.

- Conditional transfer occurs only under a control condition
- Representation of a (conditional) transfer
 - $X'T_1: A \leftarrow B$
- A binary condition ($X'T_1$ equals to 0 or 1) determines when the transfer occurs
- The content of B is transferred into A only if x is 0 and T_1 is 1



7.

Illustrate the basic symbols used in register transfer operation with examples.

Basic Symbols for Register Transfers		
Symbol	Description	Examples
Letters & numerals	Denotes a register	MAR, R2
Parenthesis ()	Denotes a part of a register	R2(0-7), R2(L)
Arrow $\leftarrow$	Denotes transfer of information	$R2 \leftarrow R1$
Comma ,	Separates two microoperations	$R2 \leftarrow R1, R1 \leftarrow R2$
Colon:	Terminates a control function	P:
Square Brackets []	Specifies an address for memory transfer	$MBR \leftarrow M [MAR]$

8.

List all possible arithmetic micro operations with an example.

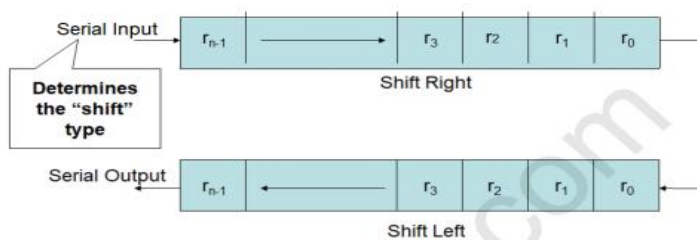
- Addition Microoperation: $R3 \leftarrow R1 + R2$
- Subtraction Microoperation: $R3 \leftarrow R1 - R2$ OR $R3 \leftarrow R1 + R2' + 1$
- One's Complement Microoperation: $R2 \leftarrow R2'$
- Two's Complement Microoperation: $R2 \leftarrow R2' + 1$
- Increment Microoperation: $R2 \leftarrow R2 + 1$
- Decrement Microoperation: $R2 \leftarrow R2 - 1$

9.

Explain logical, circular and arithmetic shift operations with example.

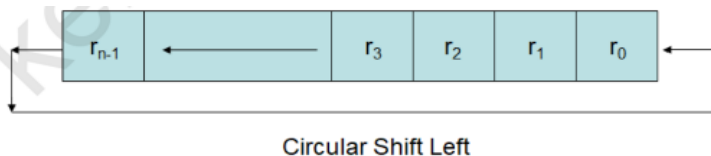
Logical Shift

- Transfers 0 through the serial input
- Logical Shift Right: $R1 \leftarrow \text{shr } R1$
The same
- Logical Shift Left: $R2 \leftarrow \text{shl } R2$
The same



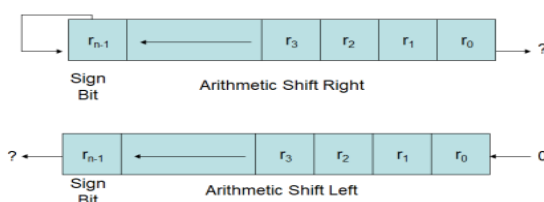
Circular Shift

- Circulates the bits of the register around the two ends without loss of information
- Circular Shift Right: $R1 \leftarrow \text{cir } R1$
The same
- Circular Shift Left: $R2 \leftarrow \text{cil } R2$
The same



Arithmetic Shift

It is also called signed shift. When shifting right with an **arithmetic right shift**, the least-significant bit is lost and the most-significant bit is *copied*.



10.

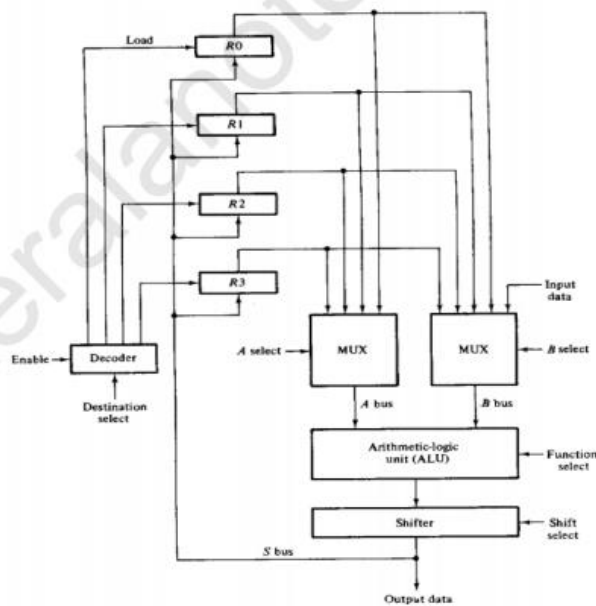
Describe various processor organization methods.

Different Processor Organization are:-

- Bus Organization
- Scratchpad Memory
- Processor with 2-port memory
- Accumulator Register

Bus Organization

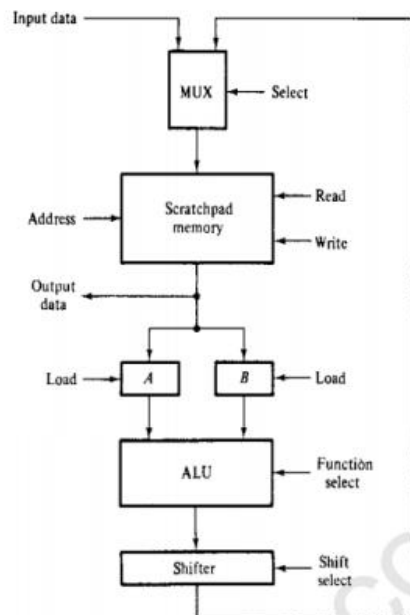
When a large number of registers are included in a processor, it is the most efficient way to connect them through common buses.



- Register connected to two MUX to form input buses A and B.
- The selection lines of each MUX select one register for the particular bus.
- The A and B buses are applied to a common ALU.
- The function selected in the ALU determines the particular operation that is to be performed.
- The shift micro-operations are implemented in the shifter.
- The result of the micro-operation goes through the output bus S into the inputs of all registers.
- The destination register that receives the information from the output bus is selected by a decoder.
- When enabled, this decoder activates one of the register load inputs to provide a transfer path between the data on the S bus and the inputs of the selected destination register.
- The output bus S provides the terminals for transferring data to an external destination.
- One input of MUX A or B can receive data from the outside
- The control unit that supervises the processor bus system directs the information flow through the ALU by selecting the various components in the unit.

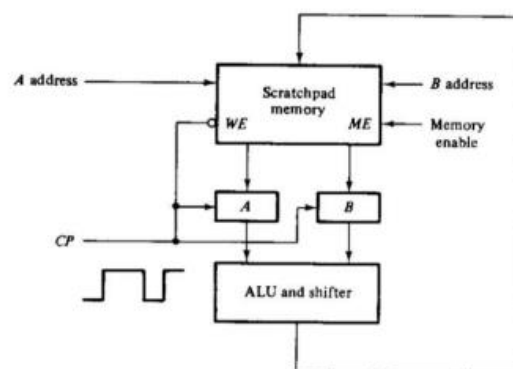
Scratch pad memory

- The register in a processor unit can be enclosed in a small memory unit.
- When included in a processor unit, a small memory is sometime called a scratchpad memory.
- The use of a small memory is a cheaper alternative to collecting processor registers through a bus system.
- The difference between the two system is the manner in which information is selected for transfer into the ALU



- Resource register is selected from memory and loaded into register A.
- A Second source register is selected from memory and loaded into register B.
- The information in A and B is manipulated in the ALU and shifter.
- Result of the operation is transferred to a memory register specifying its word address and activating the memory-write input control.
- Assume that the memory has eight words, so that an address must be specified with three bits.

Processor with 2- port memory



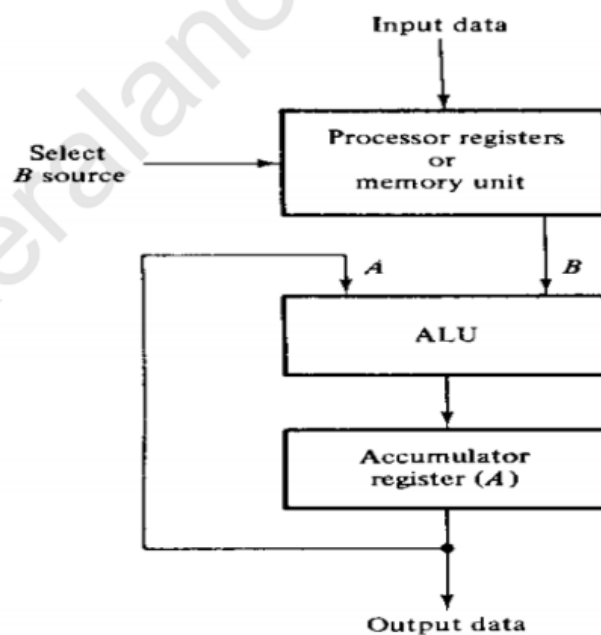
- Some processor employ a 2 port memory in order to **overcome the delay caused when reading two source registers**. A 2-port has two separate address lines to select two

words of memory simultaneously. The organization of a processor unit with a 2-port scratchpad memory is shown in figure above.

- In this way the two source register can be read at the same time.
- Memory has 2 sets of addresses. One for port A and the other for Port B. Data from any word in the memory are read into the register A by specifying an A address. Likewise data from any word in memory are read into B register by specifying a B address. The same address can be applied to the A address and B address.
- When enabled by the Memory Enable (ME) input, new data can be written into the word specified by the B address. Thus the A and B addresses specify two source registers simultaneously and the B address always specifies the destination register.
- A and B registers are latches that accept new information as the clock pulse CP is in 1-state. When CP goes to 0, the latches are disabled and hold the information that was stored when CP was 1.

Processor with Accumulator register

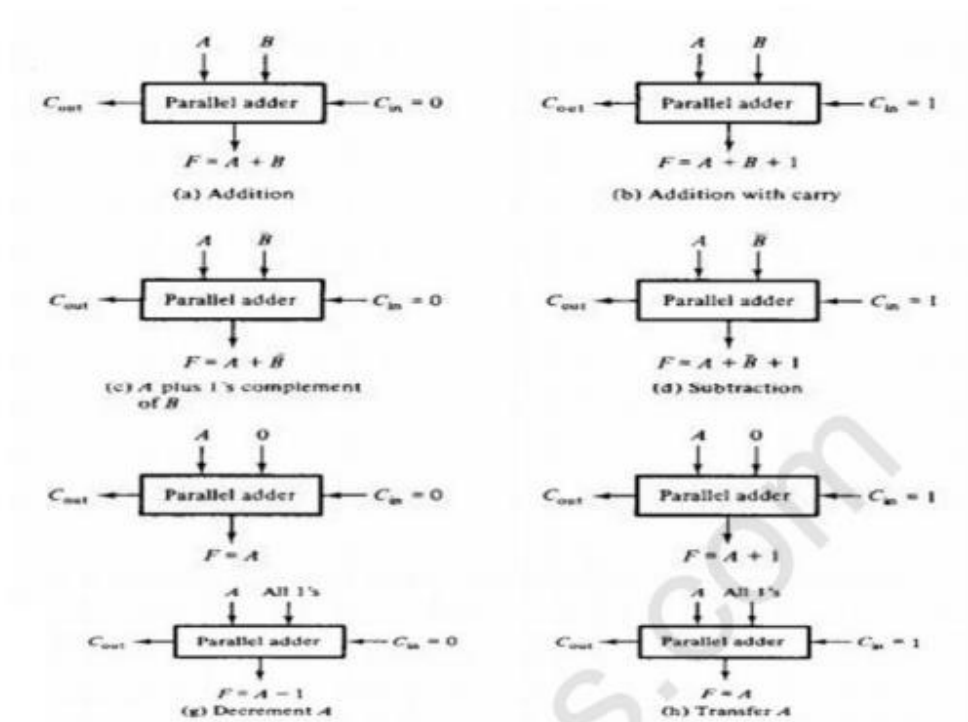
Accumulator is the intermediate storage of arithmetic and logic data in a computer's CPU



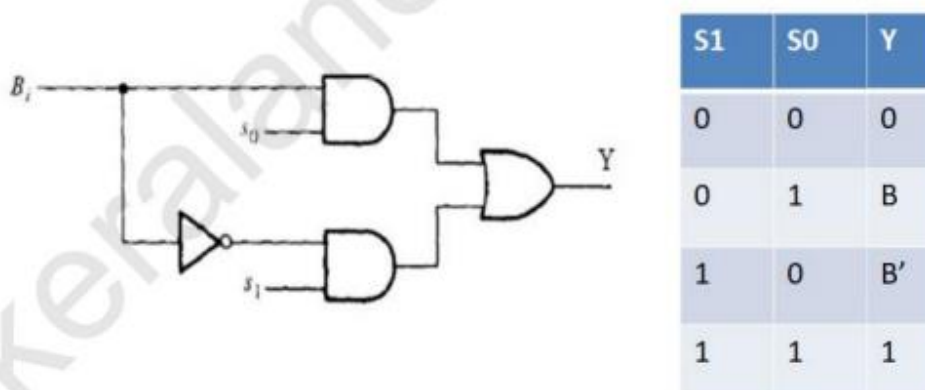
11.

Describe the design of Arithmetic circuit

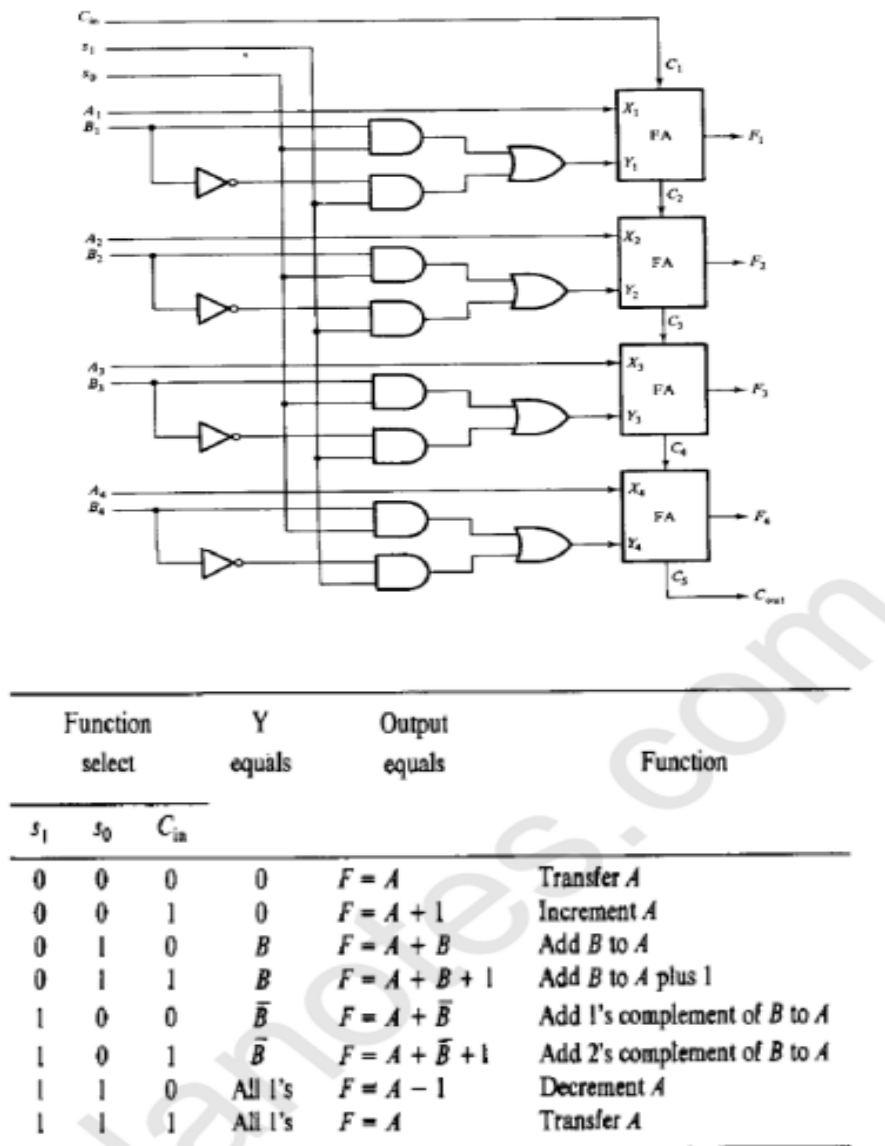
8 arithmetic operations can be described in the following figure



By changing the B input and C_{in} we get 8 operations. So the input B is applied in four different forms by using following circuit.



$$Y = S_0 B + S_1 B'$$



12.

. Explain the design of logic circuit

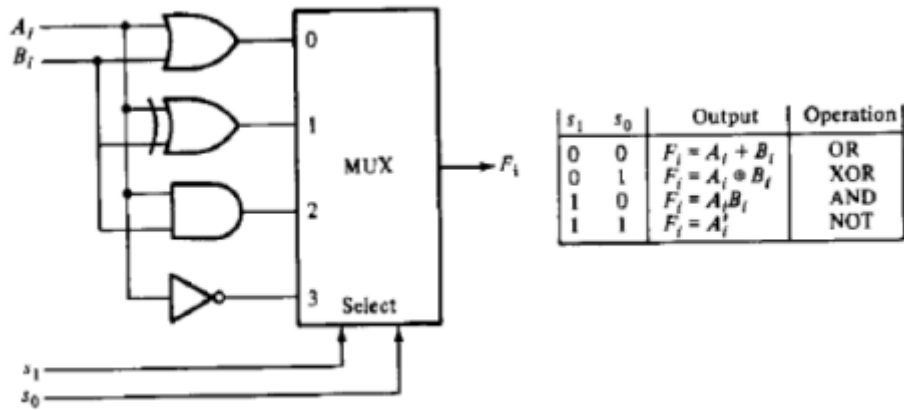
16 logic operations can be generated in one circuit and selected by means of four selection lines.

Can be obtained by means of AND, OR, and NOT (complement) operations

For three operations, we need two selection variables

But two selection lines can select among four logic operations, so we choose also the exclusive-OR (XOR) function for the logic circuit to be designed

The diagram(next slide) shows one typical stage designated by subscript i . The circuit must be repeated n times for an n -bit logic circuit



13.

. Give the function table and design of ALU

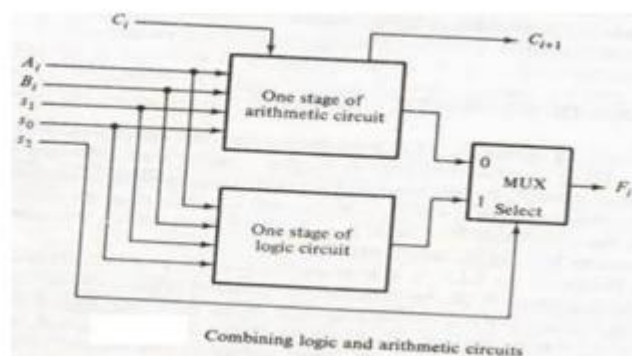
Function table

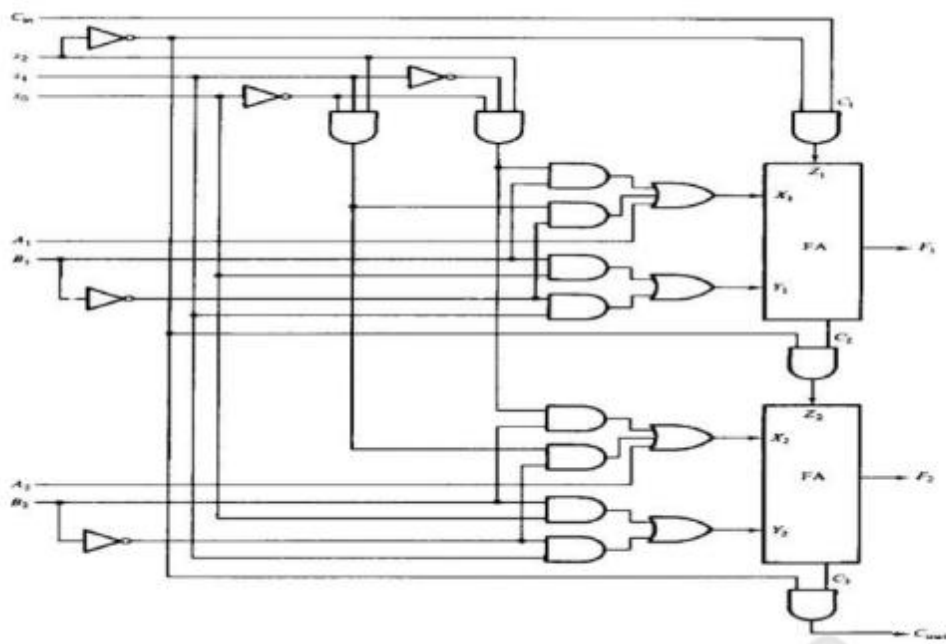
Selection				Output	Function
s_2	s_1	s_0	C_{in}		
0	0	0	0	$F = A$	Transfer A
0	0	0	1	$F = A + 1$	Increment A
0	0	1	0	$F = A + B$	Addition
0	0	1	1	$F = A + B + 1$	Add with carry
0	1	0	0	$F = A - B - 1$	Subtract with borrow
0	1	0	1	$F = A - B$	Subtraction
0	1	1	0	$F = A - 1$	Decrement A
0	1	1	1	$F = A$	Transfer A
1	0	0	X	$F = A \vee B$	OR
1	0	1	X	$F = A \oplus B$	XOR
1	1	0	X	$F = A \wedge B$	AND
1	1	1	X	$F = \bar{A}$	Complement A

Design steps

1. Design the arithmetic section independent of the logic section.
2. Determine the logic operations obtained from the arithmetic circuit in step 1, assuming that the input carries to all stages are 0.
3. Modify the arithmetic circuit to obtain the required logic operations.

One bit ALU design





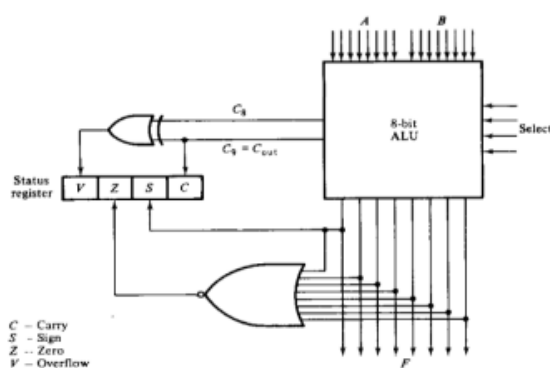
14.

Implement the status register. Give its design and function table.

Status register is a 4 bit register. The four bits are C (carry), Z (zero), S (sign) and V (overflow). These bits are set or cleared as a result of an operation performed in the ALU.

- Bit C is set if the output carry of an ALU is 1.
- Bit S is set to 1 if the highest order bit of the result in the output of the ALU is 1.
- Bit Z is set to 1 if the output of the ALU contains all 0's.
- Bit V is set if the exclusive —OR of carries C8 and C9 is 1, and cleared otherwise.

This is the condition for overflow when the numbers are in signed 2's complement representation. For an 8 bit ALU, V is set if the result is greater than 127 or less than -128.



Relation	Condition of status bits	Boolean function
$A > B$	$C = 1 \text{ and } Z = 0$	CZ'
$A \geq B$	$C = 1$	C
$A < B$	$C = 0$	C'
$A \leq B$	$C = 0 \text{ or } Z = 1$	$C' + Z$
$A = B$	$Z = 1$	Z
$A \neq B$	$Z = 0$	Z'

15.

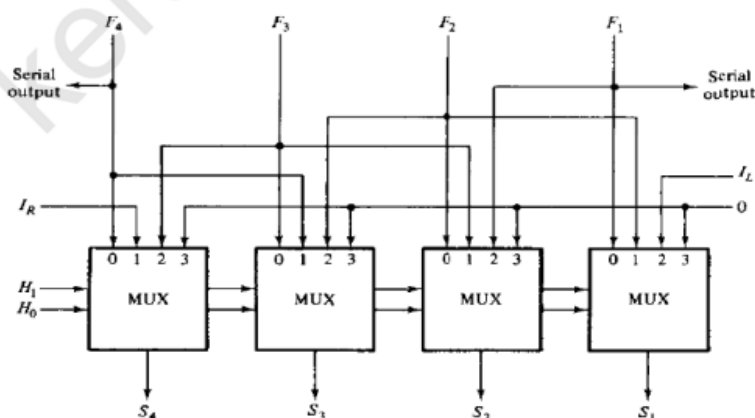
Explain the Design of a logic shifter circuit with diagram and function table.

Shifter may function in four different ways.

1. The shifter may transfer the information directly without a shift.
2. The shifter may shift the information to the right.
3. The shifter may shift the information to the left.
4. In some cases no transfer is made from ALU to the output bus.

Shifter operation can be selected by two variables H_1 H_0

- If $H_1 H_0 = 0 0$ No shift is executed and the signal from F go directly to S lines
- If $H_1 H_0 = 0 1$ Shift Right is executed
- If $H_1 H_0 = 1 0$ Shift Left is executed
- If $H_1 H_0 = 1 1$ No operations

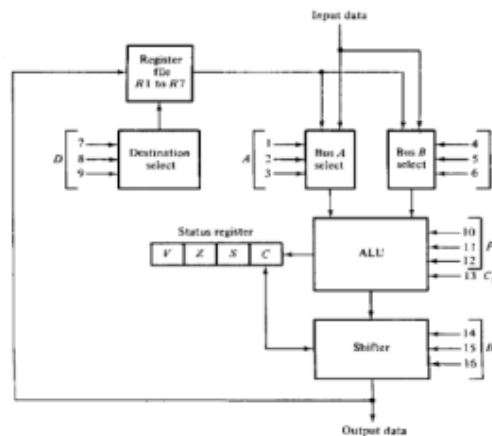


H1	H2	Operation	Function
0	0	$S \leftarrow F$	Transfer F to S
0	1	$S \leftarrow \text{Shr } F$	Shift right F into S
1	0	$S \leftarrow \text{Shl } F$	Shift left F into S
1	1	$S \leftarrow 0$	Transfer 0 into S

16.

. Design the complete Processor Unit

- The 3-bit binary code listed in the table specifies the code for each of the five fields A, B, D, input data, F, and H.
- The register selected by A, B, and D is the one whose decimal number is equivalent to the binary number in the code.
- When the A or B field is 000, the corresponding multiplexer selects the input data.
- When D = 000, no destination register is selected.
- The three bits in the F field, together with the input carry Cin, provide the 12 operations of the ALU
- There are two possibilities for F = A.
 - Carry bit C is cleared,
 - Carry bit C is set to 1.

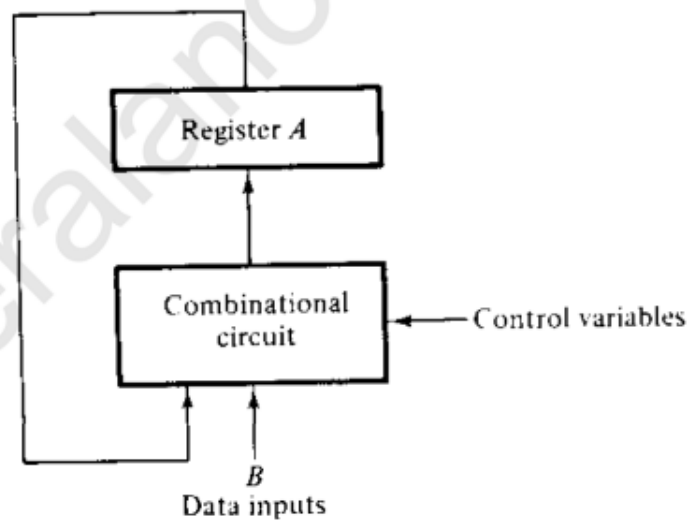


Binary code	Function of selection variables					
	A	B	D	F with $C_{in} = 0$	F with $C_{in} = 1$	H
0 0 0	Input data	Input data	None	$A, C \leftarrow 0$	$A + 1$	No shift
0 0 1	R1	R1	R1	$A + B$	$A + B + 1$	Shift-right, $I_R = 0$
0 1 0	R2	R2	R2	$A - B - 1$	$A - B$	Shift-left, $I_L = 0$
0 1 1	R3	R3	R3	$A - 1$	$A, C \leftarrow 1$	0's to output bus
1 0 0	R4	R4	R4	$A \vee B$	—	—
1 0 1	R5	R5	R5	$A \oplus B$	—	Circulate-right with C
1 1 0	R6	R6	R6	$A \wedge B$	—	Circulate-left with C
1 1 1	R7	R7	R7	$\bar{A}$	—	—

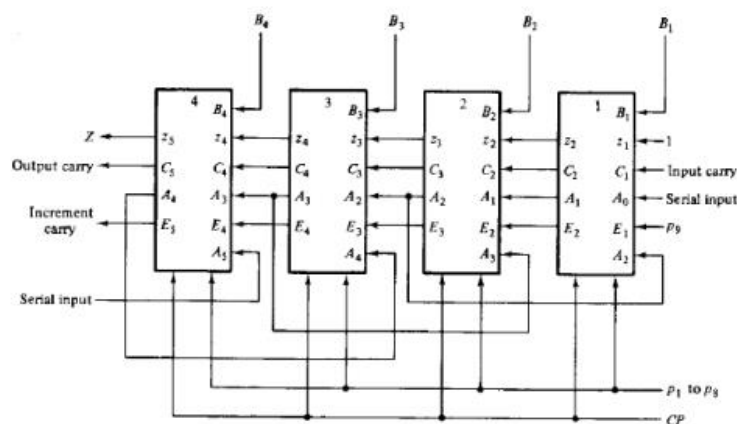
17.

Explain the Design of accumulator

- Accumulator refers to both the A register and its associated combinational circuit.
- The external inputs to the accumulator are the data inputs from B and the control variables that determine the micro operations for the register.
- The next state of register A is a function of its present state and of the external inputs.
- Accumulator can also perform data processing operations.



Control variable	Microoperation	Name
p_1	$A \leftarrow A + B$	Add
p_2	$A \leftarrow 0$	Clear
p_3	$A \leftarrow \bar{A}$	Complement
p_4	$A \leftarrow A \wedge B$	AND
p_5	$A \leftarrow A \vee B$	OR
p_6	$A \leftarrow A \oplus B$	Exclusive-OR
p_7	$A \leftarrow \text{shr } A$	Shift-right
p_8	$A \leftarrow \text{shl } A$	Shift-left
p_9	$A \leftarrow A + 1$	Increment
	If $(A = 0)$ then $(Z = 1)$	Check for zero

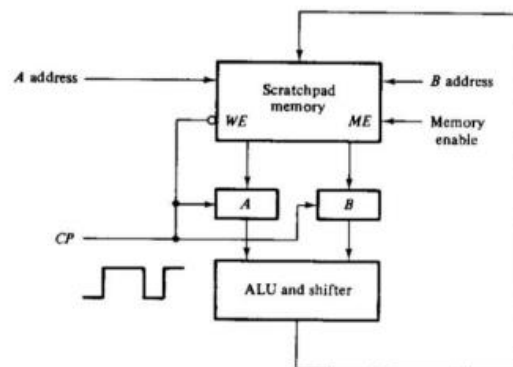


- The inputs and outputs of each stage can be connected in cascade to form a complete accumulator.
- The number on top of each block represents the bit position.
- All blocks receive 8 control variables P1 to P8 and the clock pulses from CP.
- The other six inputs and four outputs are same as with the typical stage.
- The zero detect chain is obtained by connecting the z variables in cascade, with the first block receiving a binary constant 1.
- The last stage produces the zero detect variable Z.
- Total number of terminals in the 4 bit accumulator is 25, including terminals for the A outputs.
- Incorporating two more terminals for power supply, the circuit can be enclosed within one IC package having 27 or 28 pins.
- The number of terminals for the control variable can be reduced from 9 to 4 if a decoder is inserted in the IC.
- In such cases, IC pin count is also reduced to 22 and the accumulator can be extended to 16 microoperations without adding external pins (That is, with 4 bits we can identify 16 operations).

18. How is the two port memory organization of processor unit better when compared to scratch pad memory organization?

Ans:

Processor with 2- port memory



- Some processor employ a 2 port memory in order to **overcome the delay caused when reading two source registers**. A 2-port has two separate address lines to select two

words of memory simultaneously. The organization of a processor unit with a 2-port scratchpad memory is shown in figure above.

- In this way the two source register can be read at the same time.
- Memory has 2 sets of addresses. One for port A and the other for Port B. Data from any word in the memory are read into the register A by specifying an A address. Likewise data from any word in memory are read into B register by specifying a B address. The same address can be applied to the A address and B address.
- When enabled by the Memory Enable (ME) input, new data can be written into the word specified by the B address. Thus the A and B addresses specify two source registers simultaneously and the B address always specifies the destination register.
- A and B registers are latches that accept new information as the clock pulse CP is in 1-state. When CP goes to 0, the latches are disabled and hold the information that was stored when CP was 1.

Two port memory has two address lines. Hence delay for memory access is reduced

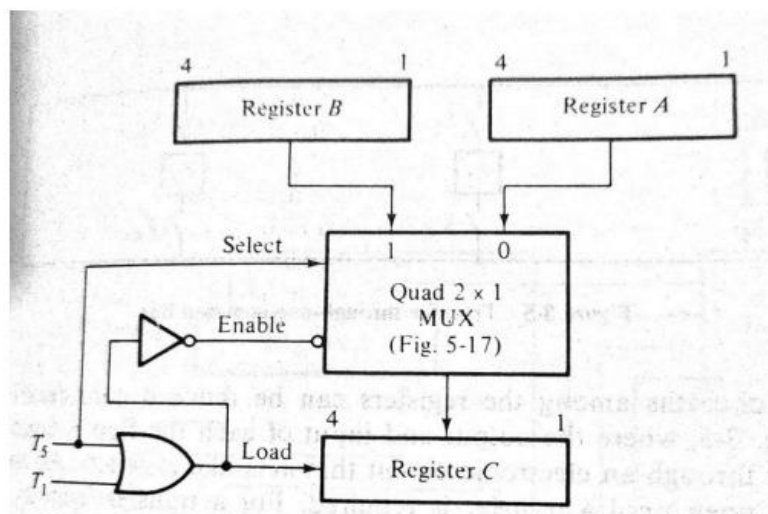
19. Give the block diagram of circuit that implements following statements in register transfer logic:

T₁: C ← A

T₅: C ← B

Ans:

Contents of A are transferred to C when timing variable T₁ occurs. Contents of B are transferred to C when timing variable T₅ occurs. A multiplexer circuit needs to be used to select between two possible paths



20.

Illustrate and explain the organization of a processor unit where processor registers and ALU are connected through common buses. Explain how the micro operation $R2 \leftarrow R3 + R4$ would be performed using this organization, where R2, R3 and R4 are processor registers.

Ans:

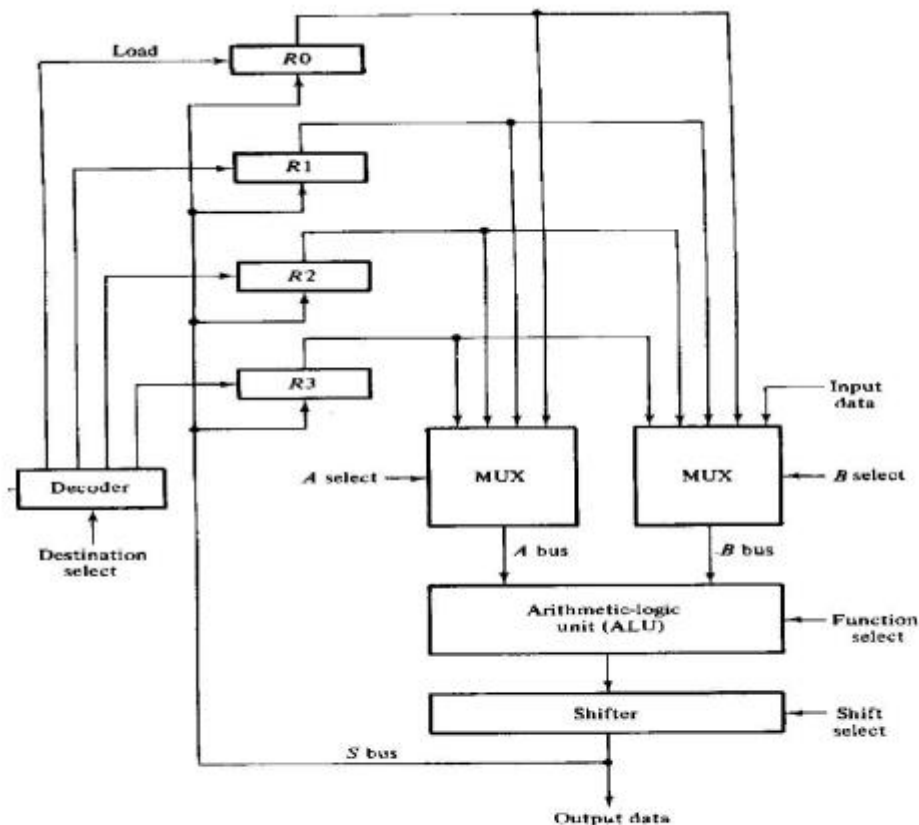


Figure 9-1 Processor registers and ALU connected through common buses

Explanation with diagram - 4 marks

- MUX A selector: Places content of R3 into Bus A.
- MUX B selector: Places content of R4 into Bus B.
- ALU function selector: Selects addition operation
- Shift selector: Selects no shift
- Decoder destination selector: Selects R2 as destination.

21.

Design 4-bit combinational logic shifter which will perform the operation given below with 2 control variable H_1 & H_0 ?

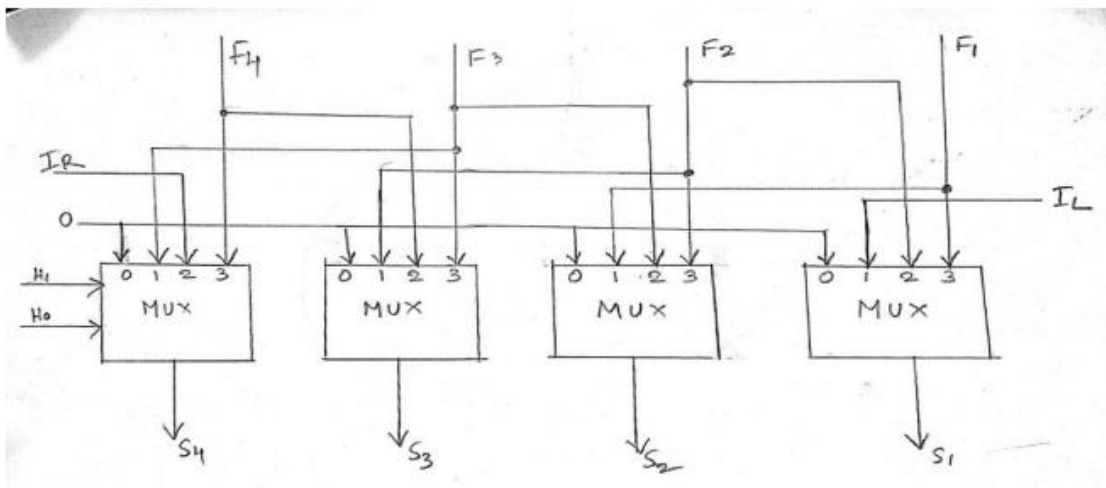
- i) Shrl
- ii) clear
- iii) Load all bits with 1

OR

Design and draw a combinational logic shifter using multiplexers with two selection variables, H_1 and H_0 . The operations of shifter should be as specified in the following table:

H_1	H_0	Operation	Function
0	0	$S \leftarrow 0$	Transfer 0's to S
0	1	$S \leftarrow \text{shl } F$	Shift left F into S
1	0	$S \leftarrow \text{shr } F$	Shift right F into S
1	1	$S \leftarrow F$	No shift

Ans:



Correct design and block diagram - 10 marks

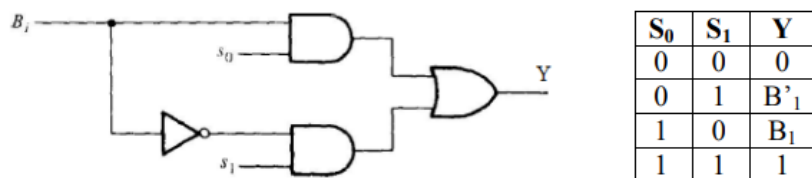
If the answer is same as that of the shifter given in textbook, 5 marks can be given

22. Draw and explain about true/complement circuit?

Ans:

The circuit that controls input B to provide the functions is called a **true/complement, one/zero** element. This circuit is illustrated in the following figure.

The 2 selection lines s_1 and s_0 control the input of each B terminal.



23. Show the hardware implementation for executing the statement $xT_1: A \leftarrow B$.

Ans:

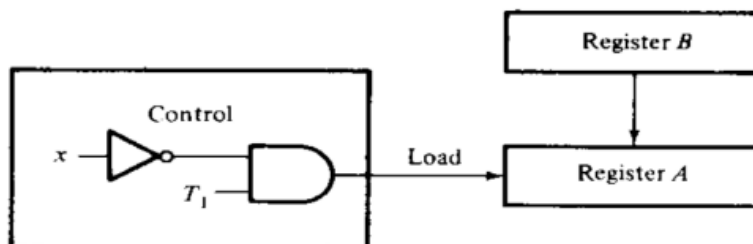
Conditional transfer occurs only under a control condition: The condition that determines when the transfer is to occur is called a **control function**. A control function is a Boolean function that can be equal to 1 or 0. The control function is included with the statement as follows

$$x'T_1: A \leftarrow B$$

The control function is terminated with a colon. It symbolizes the requirement that the transfer operation be executed by the hardware only when the Boolean function $x'T_1 = 1$, i.e., when variable $x = 0$ and timing variable $T_1 = 1$.

Hardware implementation of a controlled transfer: $x'T_1: A \leftarrow B$ is as follows

The outputs of register B are connected to the input of register A and the number of lines in this connection is equal to the number of bits in the registers. Register A must have a load control input so that



it can be enabled when the control function is 1. It is assumed that register A has an additional input that accepts continuous synchronized clock pulses. The control function is generated by means of an inverter and an AND gate. It is also assumed that the control unit that generates the timing variable T_1 is synchronized with the same clock pulses that are applied to register A. The control function stays on during one clock pulse period (when the timing variable is equal to 1) and the transfer occurs during the next transaction of a clock pulse.

24. Illustrate the design and functioning of a 4-bit arithmetic circuit within an ALU.

Explain how this circuit performs addition and subtraction operations, providing a step-by-step breakdown of the processes involved

OR

Design an adder/subtractor circuit with one selection variable s and two inputs A and B . When $s = 0$, the circuit performs $F = A + B$. When $s = 1$, the circuit performs $F = A - B$

Ans:

The design of any arithmetic circuit can be done by following the procedure outlined in the previous example.

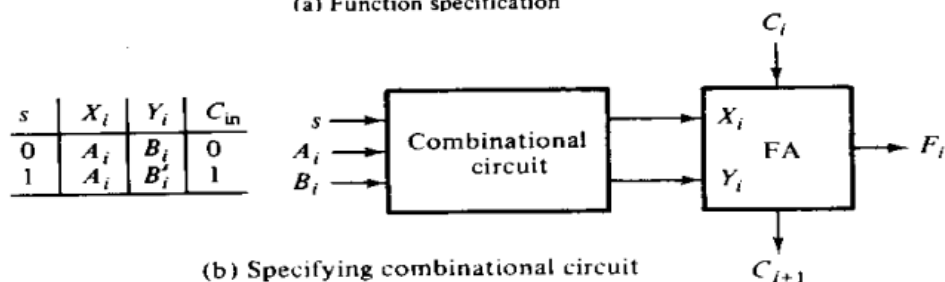
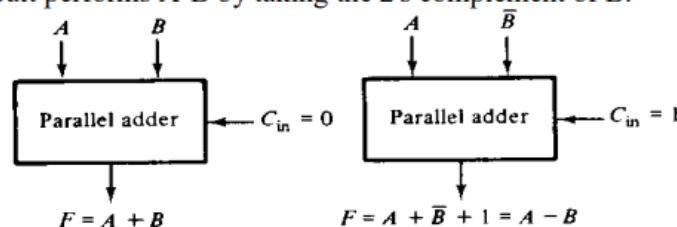
➤ Assuming that **all operations in the set can be generated through a parallel adder**

Steps in design

- Start by obtaining a **function diagram**.
- Obtain a **function table** from the function diagram that relates the inputs of the full-adder circuit to the external inputs.
- Obtain the **combinational gates** from the function table that must be added to each full-adder stage.

When $s = 0$, the circuit performs $A + B$

When $s = 1$, the circuit performs $A - B$ by taking the 2's complement of B .



s	A_i	B_i	X_i	Y_i	
0	0	0	0	0	
0	0	1	0	1	$X_i = A_i$
0	1	0	1	0	
0	1	1	1	1	$Y_i = B_i \oplus s$
1	0	0	0	1	
1	0	1	0	0	$C_{in} = s$
1	1	0	1	1	
1	1	1	1	0	

(c) Truth table and simplified equations

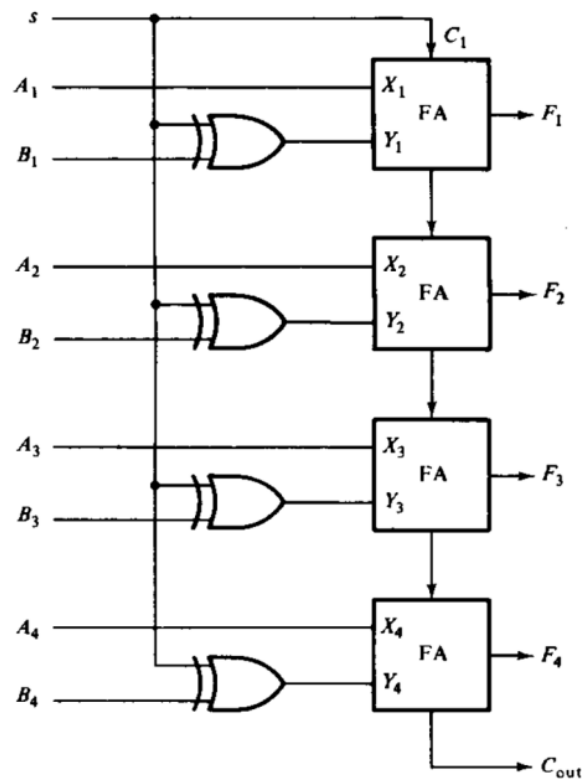


Figure 9-10 4-bit adder/subtractor circuit

25.

Explain with the help of a block diagram the design of a 4 bit status register for an 8 bit ALU. The four status bits are C (carry), S(sign), Z(zero) and V (overflow). Clearly indicate the purpose of each status bit and how they are set or reset.

OR

Give the structure of status register, which is connected to 8bit ALU ..

OR

Give the hardware design for generating status bits for an 8-bit ALU

26. Describe about arithmetic, logic and shift micro operations, listing the available operations in each category.

- 27. Write the register transfer logic format for a conditional control statement, Give an example?**
- 28. How do a binary adder circuit can be used for the implementation of logical operation AND, OR, NOT, XOR.**
- 29. Describe processor organization with diagram using i) scratchpad memory
ii) Two-port memory iii) Accumulator register**
- 30. Differentiate between arithmetic and logic micro operations in the context of register transfer logic. Provide examples for each type of operation**
- 31. What is the role of the accumulator in a processor unit? How does it contribute to arithmetic and logic operations within the system?**
- 32. What are the basic components of Register-Transfer Logic method?**
- 33. What is a scratchpad memory? Draw the block diagram of a processor employing scratchpad memory.**
- 34. What is micro-operation, With help of examples explain shift micro-operation**